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Using submerged geotextile tubes in the protection of the E. Korean shore

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Abstract

Shore erosion is currently causing millions of dollars worth of damage to shorelines and public properties not only along the east coast of Korea but also around the world. Little else needs to be said to emphasize that, without adequate protection, a very significant part of our coastline will fall prey to the ravages of the sea and to man himself. In recent years, because of the shortage of natural rock, traditional forms of river and coastal structures have become very expensive to build and maintain. Therefore, the materials used in hydraulic and coastal structures are changing from the traditional rubble and concrete systems to cheaper materials and systems. One of these alternatives employs geotextile tube technology in the construction of shore protection structures, such as groins, jetties, detached breakwaters and so on. Recently, geotextile tube technology has changed from being an alternative construction technique and, in fact, has advanced to become the most effective solution of choice.

This paper presents the various issues related to the geotextile tube construction for shore protection at Young-Jin beach on the east coast of Korea. A new approach to a stability analysis by 2-dimensional limit equilibrium theory is highlighted and the hydraulic model test results and case history of Young-Jin beach projects are described. Based on the results of stability analysis and hydraulic model tests, a two line geotextile tube installed with zero water depth above crest was found to be more stable and effective for wave absorption than other design plans. Also, the shoreline at Young-Jin beach was extended by about 2.4–7.6 m seaward, and seabed sand was gradually accumulated around areas covered by the geotextile tube.
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Keywords: Shore protection; Breakwater; Geotextile tube; Hydraulic model test; Field monitoring; Stability

1. Introduction

In recent years, traditional forms of river and coastal structures have become very expensive to build and maintain, because of the shortage of natural rock. As a consequence, the materials used in hydraulic and coastal structures are changing from traditional rubble and concrete systems to cheaper materials and systems such as gabion, slags, geosynthetics, and so on. Moreover, shorelines are being continually eroded by the sea wave action, and the river and coastal structures are frequently damaged by both anthropogenic and natural causes such as overwash, and storm.

Geosynthetics are being increasingly used in civil and environmental applications. One of these applications is the use of geotextile tube technology. Geotextile tubes, hydraulically or mechanically filled with dredged materials, have been variously applied in hydraulic and coastal engineering fields. The geotextile tube technology is mainly used for flood and water control, but they are also used to prevent beach erosion and, for shore protection. Woven, non-woven, and composite synthetic fabrics, i.e. geotextile, have been used for the past 30 years for various types of containers, such as small hand-filled sandbags, 3-dimensional fabric forms for concrete paste, large soil and aggregate filled geotextile gabion, prefabricated hydraulically-filled containers, and other innovative systems involving containment of soils using geotextile. Koerner and Welsh (1980), and Pilarczyk (1990, 1995) provide an overview of the many primarily erosion control applications using the various types of containers. Heibaum (2002) also presented various case

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histories of geosynthetic containers applied as armour, ballast, filter, storage, core for hydraulic structures, flood protection, scour repair and protection, and improvement of earth dam. Sprague (1995) presented the basic design concepts for geotextile tubes filled with dredged material. The geotextile sheets are permeable, yet soil-tight, so that any excess water drains from the geotextile tube. This causes the tube height to decrease, so that the tube may have to be pumped more than once in order to achieve the desired height (Leshchinsky, 1993). There are inlets at the upper part of the tube where the pumping



Fig. 1. The location of our experimental site on the east coast of Korea.



Fig. 2. Erosion phenomena of east coast of Korea (Young-Jin beach).

hose is inserted. The number and interval of inlets is dependent upon the type of soil being used. Typical lengths and widths of geotextile tubes are 150, 180 m and 45 m, respectively, with the effective height of 1.5, 2.0 m. The interval of inlet is shorter for sandy soil and longer for the case of clayey soil (Leshchinsky et al., 1996).

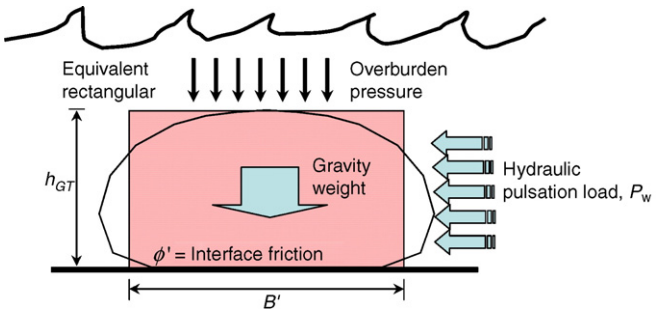


Fig. 3. Schematic diagram of 2-dimensional hydraulic stability.

Recently, geotextile tubes filled with dredged material have been used in dike and breakwater construction for a number of projects around the world, and their use in this field is growing very fast. Some of the most attractive advantages of geotextile tube technology is that it can be used for in-situ filling materials by hydraulic pumping; it can be also implemented with lower costs and faster construction than other technology. Because of the lower price and easier installation, geotextile tube systems can be good alternatives for hydraulic and coastal structures. Dikes and levees are among the primary uses of geotextile tubes. Dikes can be constructed up to 2.0 m tall to provide flood protection. By stacking the tubes, an even greater height can be achieved. These tubes can also be attached to the top of a floodwall to provide greater flood control (Perry, 1993). In Germany, a 15 km dike of sand-filled geotextile tubes was constructed in Leybucht. This system proved to be a very efficient and durable method of water control (De Bruin and Loos, 1995). Groins can be very effective when used for shoreline protection. Sand-filled geotextile bags are a very reasonable alternative to other groin types. Sand-filled bags can be also be used for revetments or bulkhead protection (Gutman, 1979). Environmental dredging and backfill technology using geotextile tubes were reported by Fowler et al. (1995, 2002) and Mori et al. (2002).

For geotextile tubes, the major design considerations are related to the integrity of the units during release and impact, and

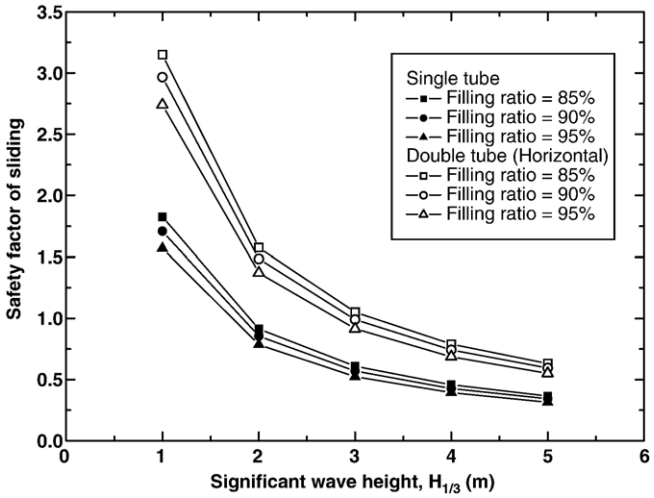


Fig. 4. Safety factor of sliding.

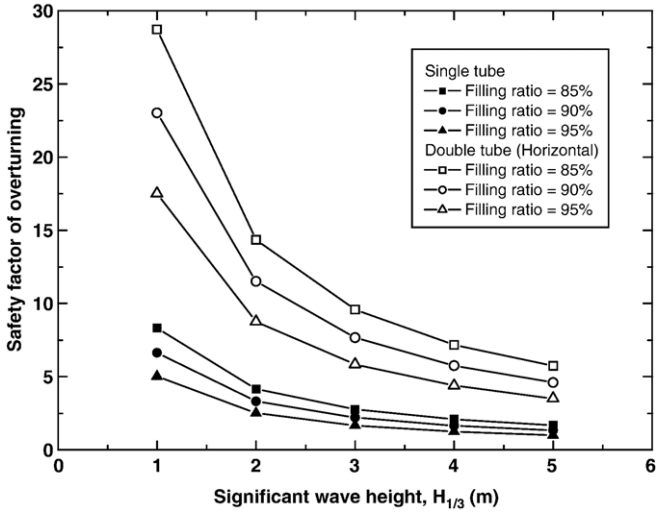


Fig. 5. Safety factor of overturning.

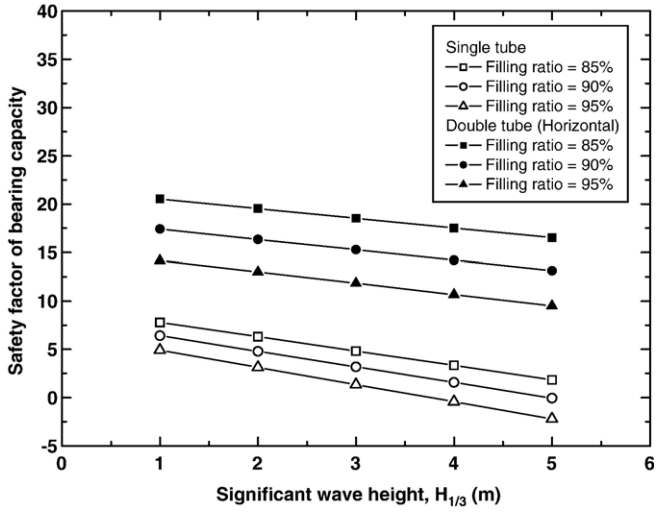


Fig. 6. Safety factor of bearing capacity failure.

the accuracy of placement and the stability under current and wave attack (Pilarczyk, 1998, 2000). The following design aspects are also important; (1) consistency of the filling material, (2) retention capacity of geotextile, (3) strength of geotextile, (4) hydrodynamic stability. The main advantage of geotextile tube systems in comparison with more traditional methods (rock, prefabricated concrete unit, block mats, asphalt, etc) are reduction in work volume, execution time, and cost, the use of local material, low-skilled labor and locally available equipment.

Table 1
Model-prototype scale relations (1:50 scale)

Characteristic	Dimension	Scale relations model : prototype
Length	l	l_r=1:50
Area	l^2	a_r=1:2500
Volume	l^3	v_r=1:125000
Time	$l^{1/2}$	t_r=1:7.07

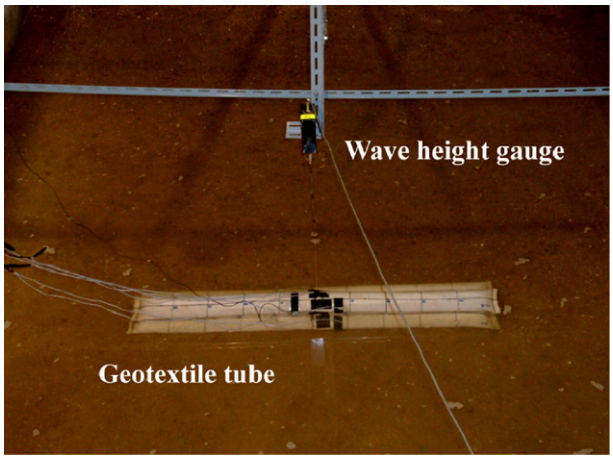


Fig. 7. Model unit of geotextile tube.

The objective of this paper is to examine several issues associated with hydrodynamic behavior of submerged geotextile tubes such as hydraulic stability, wave absorption capacity during wave attack, and also the feasibility of geotextile tube technology for shore protection during and after construction. This paper proposes a hydrodynamic stability analysis method based on 2-dimensional limit equilibrium theory and presents some of the experimental data that verify its feasibility for submerged geotextile tubes. Following is a review of studies for prediction of hydrodynamic characteristic of submerged geotextile tube, especially those dealing with prediction based upon the hydraulic model tests. The final and, main portion of this paper, covers the construction procedure and the results of in-situ measurement such as effective height, vertical stress at the bottom of the geotextile tube, shoreline variation, and water depth of the near shore area.

2. Site description

The East Sea is a marginal sea adjoining the North Pacific through the Korean Strait in the south, and through the Tsugaru, Soya and Tartar Straits in the north. The East Sea forms a rectangular pattern with a total area of 1.008×10^6 and a mean depth of 1684 m making them wider and deeper than those of Yellow Sea, the South Sea of Korea, and the Korean Strait.

Shore erosion is currently causing severe damage to shoreline scenic views and to public property along the east coast of Korea (Figs. 1 and 2). Shore erosion is caused by the energy of wave

Table 2
Hydraulic model tests

Cross section	Water depth above tube crest	Significant wave height (m)
Single geotextile tube	Zero water depth above crest	3.0
	0.5 H_{GT} water depth above crest	4.0
Two lines geotextile tube	Zero water depth above crest	5.0
	0.5 H_{GT} water depth above crest	6.0

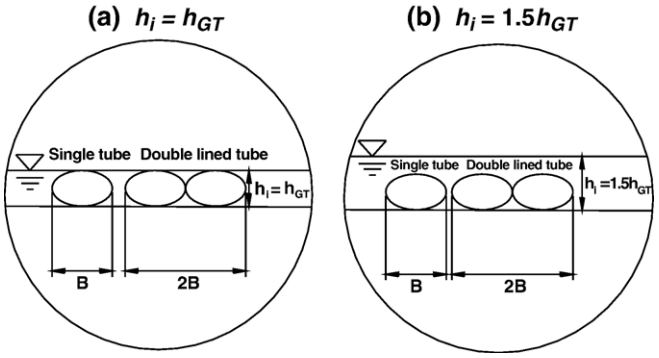


Fig. 8. Geotextile tubes under test conditions.

attack, periodic tides, and currents which are produced by seasonal winds. Annual/interannual variabilities in strength and transport of East Korea Warm Current (EKWC) and North Korea Cold Current (NKCC) dominate the circulation patterns of upper waters in the Ullung Basin. These currents are closely associated with annual episodes of the fluctuation in separation latitude at the Korea coast, the formation of mesoscale eddies, and the southward intrusion by the NKCC. The process of shoreline erosion on the east coast of Korea is most severe when the significant wave heights are varied up to 4.0 m range in winter season. The severe erosion process continues even after early spring season during 4 months, as the shoreline have become stiffer and high energy wave from current and storm can still attack them. In order to reduce the erosion damage to beaches and shoreline, the Ministry of Marine Affairs and Fisheries (MMAF) of Korea must apply shore protection methods every year. Effective methods of shore protection are designed to slow or stop the erosion process and erosion control by dissipating wave energy and/or preventing shoreline attack. Therefore, the decision-makers of many cities along the east coast of Korea have applied various types of shore protection method over the years. In spite of the fact that the choice of shore protection method to be used at each site was based upon accurate design criteria, most of city and province officials relied on only one type of material and structure. The structure type is the near shore breakwater and the material for its implementation was semi-interlocking precast concrete segments (tetrapod). However, this technique requires provision for an appropriate foundation to support the super-structure, including a toe protection which will remain stable under design wave forces. Relatively heavy duty equipment is required for breakwater construction and it is apparent that construction practices and procedure are difficult and involved as compared to construction of similar structures on the shore zone. Considerable attention has been given to the use of concrete segment type breakwaters to attenuate wave energy.

Therefore, the city and province officials considered other alternative solution to find more inexpensive and environmental sustainable technology and geotextile tube technology was implemented was Young-Jin beach on the east coast of Korea (see Fig. 1).

3. Hydraulic stability analysis of geotextile tube

The main problem in geotextile tube technology is lack of proper design criteria such as hydraulic stability, structural functionality and knowledge of their behavior during and after construction. In the past, the design of these systems was mostly based on rather vague experience than on general valid calculational methods. More research, especially concerning large-scale tests and the evaluation of the performance of projects already realized, is still needed. Recently, new preliminary design criteria supported by model and prototype tests, and some stability analysis calculations have been studied. In general, the hydraulic design criteria are mainly composed of hydraulic stability analysis and structural function before/after the construction of geotextile tubes. In this paper, the hydraulic stability analysis based on 2-dimensional limit equilibrium theory is briefly described.

3.1. 2-dimensional hydraulic stability analysis

Several causes are responsible for the failure of geotextile tube structures. These causes including sliding, overturning, bearing capacity failure, and migration of the sand in the tube, forcing associated with waves (including breaking waves), non-breaking waves, and waves that propagate over the tube. In order to assess the stability of the filled geotextile tube structure, current wave forces have to be estimated. Though a definitive analysis technique has not been established, a modified Minikin approach, as outlined in the U.S. Army Corps of Engineers' Shoreline Protection Manual, may provide a reasonable approach to assess the stability of filled units under wave loading.

In this paper, the theoretical stability analysis employed is a 2-dimensional hydraulic stability analysis, based on the linear wave theory and geotechnical stability analysis method. Several methods that can be pursued to address impact loading are

Table 3
Wave conditions used in testing hydraulic models

Prototype wave condition		Model wave condition	
Significant wave height (m)	Period (s)	Significant wave height (cm)	Period (s)
3	6.69	6	0.95
4	7.72	8	1.09
5	8.63	10	1.22
6	9.46	12	1.34

Table 4
Specifications of the hydraulic model testing device

Item	Specification
Channel size	Width 7.0 m, length 30.0 m, depth 1.5 m
Wave plate size	Width ca. 7.0 m, height ca. 1.4 m
Max. water depth	1.0 m
Max. wave height	0.3 m
Period	0.5–5.0 s
Control type	Active wave absorber control system
Wave type	Regular and irregular

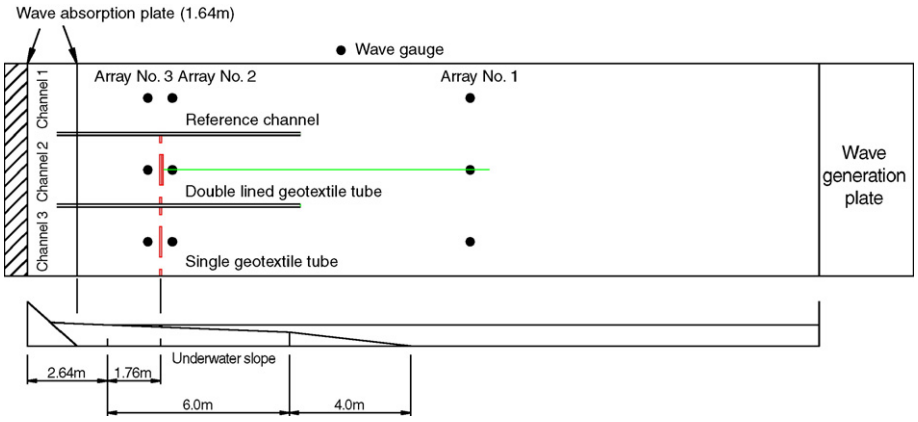


Fig. 9. Schematic diagram of hydraulic model test.

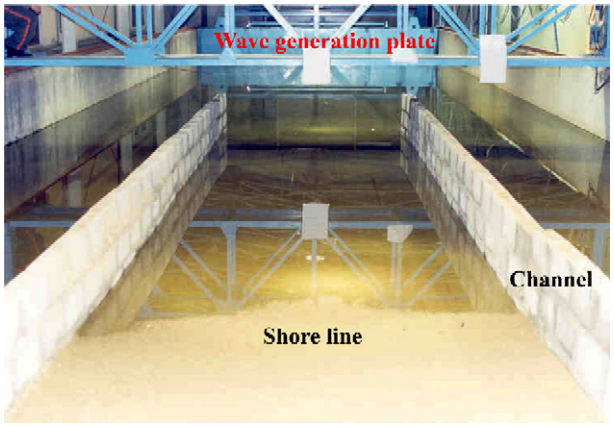


Fig. 10. Hydraulic model test device and flow channel.

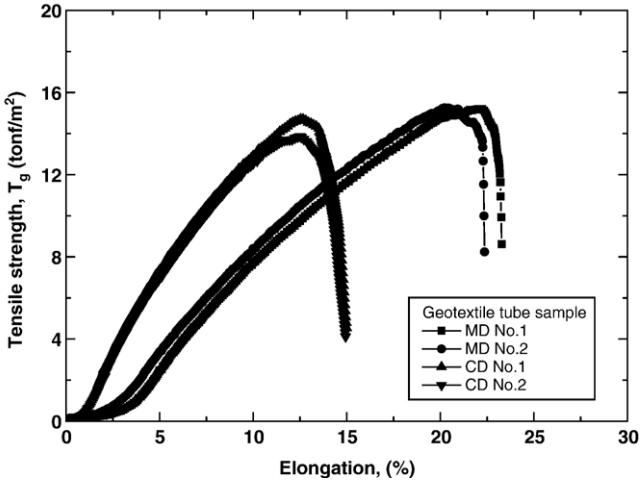


Fig. 11. Tensile strength test results.

addressed by Hiroi (1920), Minikin (1963), Honma and Horikawa (1965), and Goda (1985). Hiroi's equation is particularly widely used in Japan and the Asian region. The hydrodynamic pulsating load to a geotextile tube is evaluated by Hiroi's empirical equation as written as;

$$P_w = 1.5 \times \rho_0 \times H_{1/3} \quad (1)$$

where, P_w = hydrodynamic pulsating load, ρ_0 = unit weight of sea water, $H_{1/3}$ = significant wave height. On the other hand, the resistance of a geotextile tube structure to these various loading factors include gravity, friction (between the fabric and the bedding it sits on), foundation and fabric strength.

Table 5
Physical properties of textile used in geotextile tube construction

Physical properties	Test method	Unit	Min. value
Material, weight	ASTM D-5261	Oz/yd^2	PP, 20.0
Tensile strength	ASTM D-4632, 4595	kN/m	175.0
Elongation	ASTM D-4632, 4595	%	18.0
Tear strength	ASTM D-4533	kN/m	70.0
Seam strength	ASTM D-4884	kN/m	105.0
Equivalent opening size	ASTM D-4751	US Sieve	# 100
Permittivity	ASTM D-4491	s	0.10
Ultraviolet degradation (percent strength retained @ 500 h)	ASTM D-4355	–	70%

The following discussion will focus on the safety factor of a geotextile tube against external loading and the schematic diagram of 2-D stability analysis is shown in Fig. 3. To simplify the 2-dimensional analysis, the gravity weight of geotextile tube was calculated by equivalent rectangular shape in the effective height. The factor of safety against sliding can be expressed by Eq. (2).

$$SF_{\text{sliding}} = \frac{F}{P_h} = \frac{P_v \times \tan\phi'}{P_w \times h_{GT}} \quad (2)$$

in which P_h = horizontal force, F = vertical force, P_v = overburden pressure and gravity weight of geotextile tube, P_w = hydrodynamic

Table 6
Physical properties of filling material

Item	Quantity
Specific gravity, G_s	2.64
Effective size, D_{10}(mm)	0.50
Uniformity coefficient, C_u	1.90
Coefficient of gradation, C_c	0.84
Max. dry unit weight, $\gamma_{d(max)}$(kN/m^3)	16.38
Optimum moisture content, w_{opt} (%)	14.2
Interface friction angle(dry) (°)	34.7
Internal friction angle(wet) (°)	32.0
USCS	SP

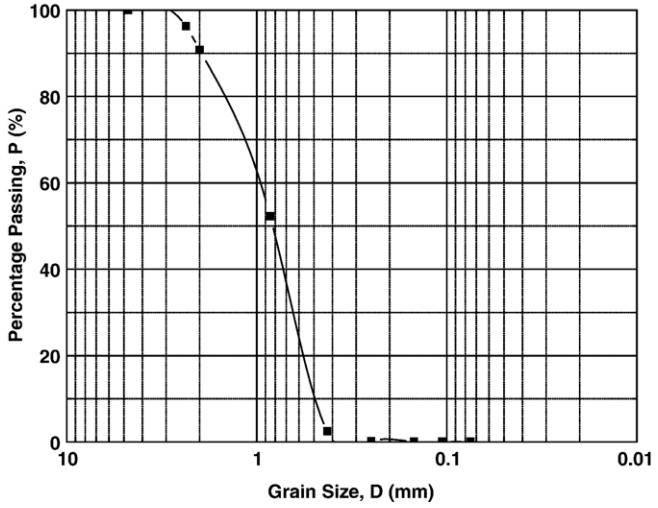


Fig. 12. Grain-size distribution curve of filling material.

pulsating load, h_{GT} = effective height, ϕ' = interface friction angle between geotextile and base sand. The factor of safety against overturning about the toe of an equivalent rectangularly shaped tube can be expressed by Eq. (3).

$$SF_{\text{overturning}} = \frac{M_R}{M_O} = \frac{P_v \times \frac{B'}{2}}{P_w \times \frac{h_{GT}}{2}} \tag{3}$$

in which B' is width of an equivalent rectangularly shaped tube. The overburden pressure and gravity weight as transmitted to the base soil should be checked against the ultimate bearing capacity of base soil. The factor of safety against bearing capacity failure can be determined by Eq. (4).

$$SF_{\text{bearing capacity}} = \frac{Q_u}{Q_a} = \frac{cN_c + \left(\frac{1}{2}\right)\gamma_s B' N_\gamma}{\frac{P_w}{B'-2e'}} \tag{4}$$

in which c is cohesion of the base soil, N_c,N_γ = bearing capacity factors by the internal friction angle of saturated base soil, γ_s = submerged unit weight of base soil, e'=the eccentricity of the hydrodynamic pulsating load as shown in Eq. (5).

$$e' = \frac{P_w \times h_{GT}}{3F} \tag{5}$$

3.2. Stability analysis results

Two-dimensional stability analysis results are shown in Figs. 4, 5 and 6. The geometry of the geotextile tube, the characteristics of the filling material, and the base soil properties are the same as the field conditions at Young-Jin beach. The variable parameters of the stability analysis are the filling ratio and the significant wave height, they, too, are varied from 85–95%, and 1.0–5.0 m, respectively. From the results of the 2-dimensional stability analysis, the safety factor was decreased on increasing the filling ratio and significant wave height. The safety factor of bearing capacity is varies linearly with filling ratio and significant wave height. However, the safety factors of sliding and overturning exponentially decay because of the relationship of equivalent width, effective height, and significant wave height. Also, the two lines tube is more stable than single installed tube. Based on the 2-D limit equilibrium analysis, a single installed geotextile tube at Young-Jin beach will be unstable against the significant wave height of more than ca. 2.0 m.

4. Hydraulic model test of geotextile tube structure

4.1. The prototype

The Young-Jin beach is located on the east coast of Korea at Young-Jin Ri, approximately 1.2 km south of Jumun-Jin harbor and 18 km north of Gangneung city. The shoreline of this area consists of broken irregular cliffs about 7 to 10 m high with numerous rocks extending several hundred meters offshore. Small beaches front heads of coves in the immediate vicinity. The Young-Jin beach is approximately 50 m wide and 700 m long at Young-Jin cove. Young-Jin beach is open to the East Sea and is exposed to high waves generated by currents and storms. Waves in excess of 4.0 m (significant wave height) approach the cove from northwest to southwest during the winter season. The Young-Jin beach has been gradually eroded by current and wave attack over the last few years. The wharf facility and scenic view road has also experienced strong surging problems due to high wave energy, resulting in damage to the smaller and lighter wharf at Young-Jin harbor. Shore protection and erosion control of Young-Jin beach would result in the reduction of shoreline and scenic view road damages, increased tourism and commercial shell catch, and a reduction in operation and maintenance costs for the existing Young-Jin harbor facilities and road system. Also, project construction would employ local construction companies and enhance area redevelopment, thereby contributing to the local economic base.

Local residents expressed interest in the construction of a geotextile tube structure that would provide shoreline protection from the extreme wave conditions at Young-Jin cove. To protect the shoreline of Young-Jin beach while at the same time retaining the beautiful beach view, it has been decided to construct a submerged geotextile tube structure (detached breakwater). At the request of the Department of Construction of Gangneung city

Table 7
Results of stability tests (N.D.: No Displacement)

	Zero water depth above crest				0.5 H water depth above crest			
Significant wave height	3.0 m	4.0 m	5.0 m	6.0 m	3.0 m	4.0 m	5.0 m	6.0 m
Two lines tube	N.D.	N.D.	N.D.	0.75 cm	N.D.	0.55 cm	0.77 cm	0.85 cm
Single tube	N.D.	N.D	N.D.	10.84 cm	N.D.	0.72 cm	1.01 cm	12.76 cm

(DCG), 2-dimensional wave tests were conducted at the Kunsan Ocean and Hydraulic Research Center to determine the optimum geotextile tube cross section and head in terms of stability and wave transmission. The initial purpose of the investigation was to determine, by 2-dimensional flume tests, the stability response of four installation conditions for the proposed geotextile tube structure and to evaluate the overall performance relative to wave transmission. Finally, based on the results of the initial tests

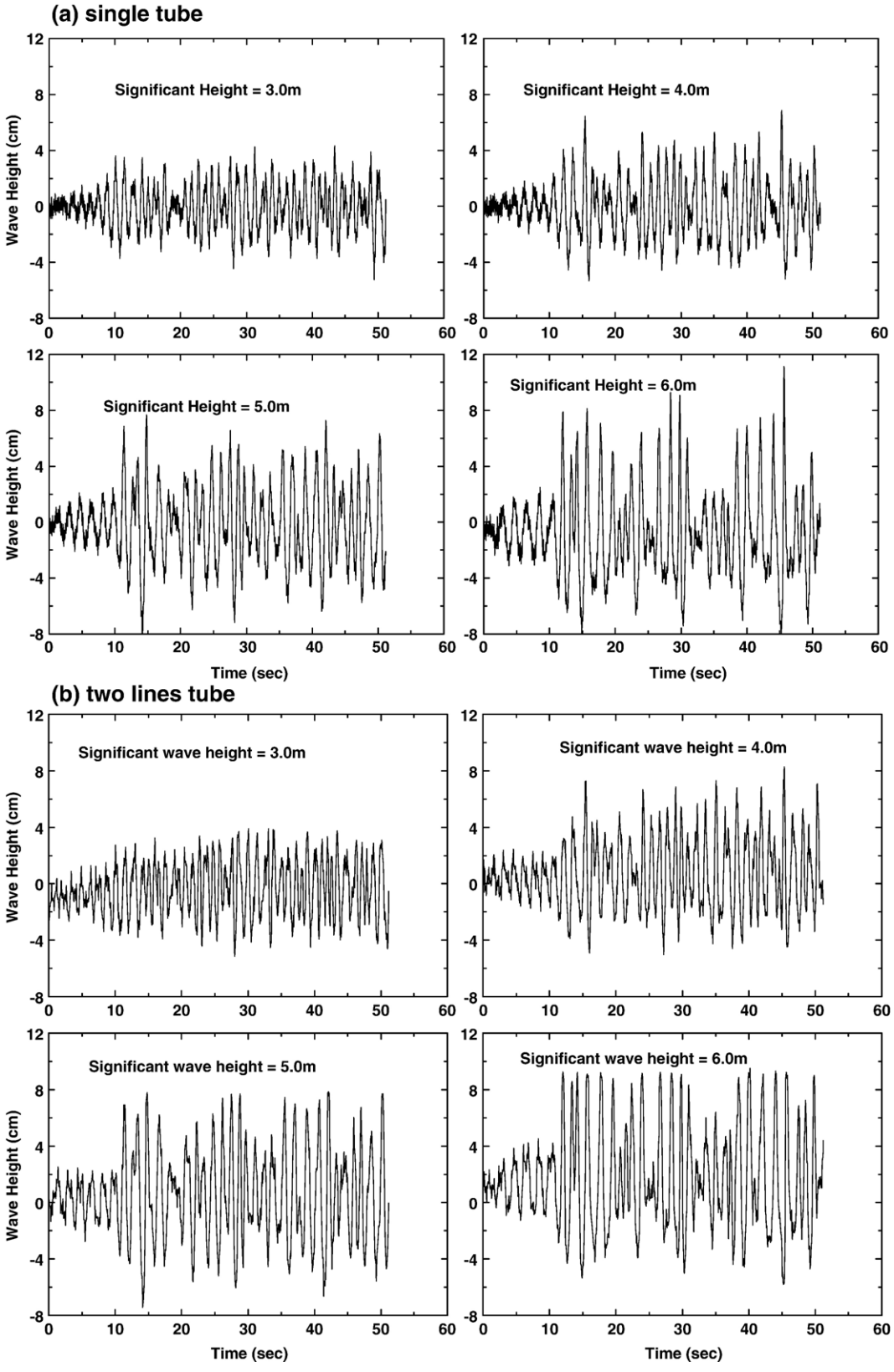


Fig. 13. Incident wave profiles.

and economic considerations, the best cross section alternative would be selected. This paper describes the design, facilities used, and results of the 2-dimensional stability and the laboratory wave transmission tests.

4.2. The model

2-dimensional stability and laboratory wave transmission tests were conducted at a geometrically undistorted linear scale

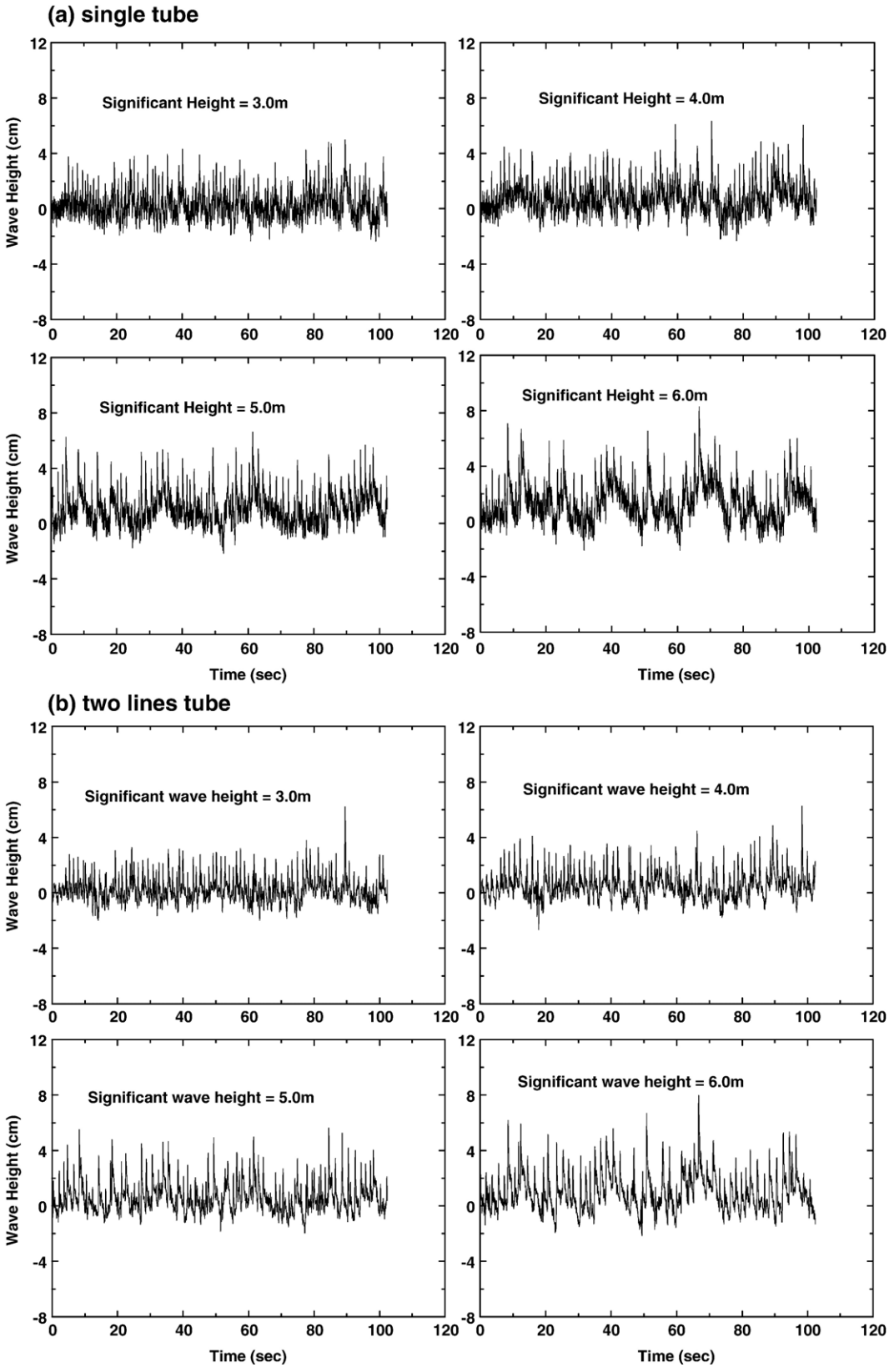


Fig. 14. Transmitted wave profiles (zero water depth above crest).

of 1:50, model to prototype. Scale was based on the availability of a suitably sized model geotextile tube and the capabilities of the available wave generator to produce the required wave heights at the modeled water depths. Time relations were scaled according to the Froude model (Stevens et al., 1942). Model to prototype relations were derived in terms of length l and time t

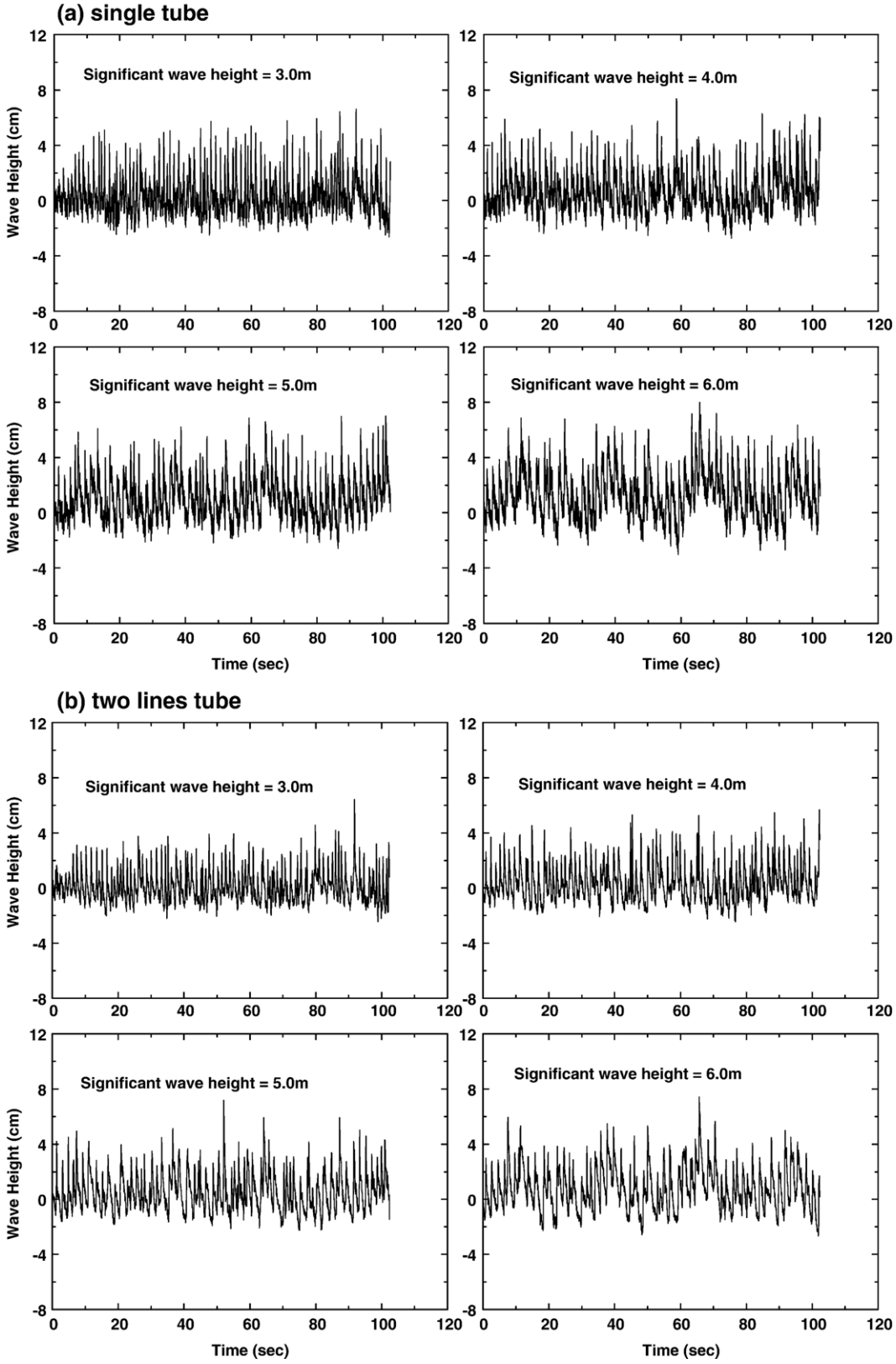


Fig. 15. Transmitted wave profiles (0.5 H water depth above crest).

Table 8
Transmit wave height and ratio

Significant wave height, (m)	Incident wave height (cm)	Transmitted wave height (Ratio)	
		Zero water depth above crest	0.5 H water depth above crest
3.0	6.0 (two lines)	3.41 (56.83%)	4.35 (72.50%)
	6.0 (single)	3.71 (61.83%)	4.64 (77.33%)
4.0	8.0 (two lines)	3.81 (47.63%)	4.95 (61.88%)
	8.0 (single)	3.94 (49.25%)	4.90 (61.25%)
5.0	10.0 (two lines)	4.25 (42.50%)	5.57 (55.70%)
	10.0 (single)	4.41 (44.41%)	5.48 (54.50%)
6.0	12.0 (two lines)	5.38 (44.83%)	6.88 (57.33%)
	12.0 (single)	5.55 (46.25%)	6.50 (54.17%)

shown in Table 1. The model units used in 2-dimensional stability and laboratory wave transmission tests are shown in Fig. 7. In a hydraulic model investigation of this type, gravitational forces predominate. A practical basis for similarity in models in which gravitational forces predominate is the equating of inertia/gravity force ratios of model and prototype as shown in Eqs. (6)–(8).

$$\left(\frac{F_i}{F_g}\right)_m = \left(\frac{F_i}{F_g}\right)_p \tag{6}$$

$$\left(\frac{\rho V^2 L^2}{\rho g L^3}\right)_m = \left(\frac{\rho V^2 L^2}{\rho g L^3}\right)_p \tag{7}$$

$$\left(\frac{V^2}{gL}\right)_m = \left(\frac{V^2}{gL}\right)_p \tag{8}$$

In which $\frac{V}{\sqrt{gL}}$= Froude number, F_i = inertia force= $\rho V^2 L^2$, F_g = gravity force = $\rho g L^3$, ρ is density, L is length, V is velocity, g is local acceleration due to gravity, m is subscript of model system, p is subscript of prototype system.

The laboratory hydraulic model tests were performed to estimate the hydrodynamic behavior of geotextile tubes as a function of the cross section, water depth above crest and significant wave height. The detail test series are shown in Table 2 and Fig. 8. Construction of the modeled section simulated prototype construction as closely as possible. The geometrical geotextile tube condition of the prototype was 1.8 m of effective height, 50.0 m of total length, not connected each tubes, and the filling ratio of geotextile tube was fixed at 85%. The geometrical similarity was 1:50. The wave conditions of hydraulic model tests by similarity are described in Table 3. The applied spectrum of significant wave height was of the Bretschneider–Mitsuyasu type. Hydraulic model tests were conducted simultaneously for single and two lines geotextile tube structures at each individual flow flume.

4.3. Test facilities and procedure

Hydraulic model testing was conducted in the Kunsan Ocean and Hydraulic Research Center at the Kunsan National University, Korea. The free surface channel had a width of 7.0 m, length 30.0 m, and a maximum water depth of 1.0 m. The specifications of the hydraulic model testing device are tabulated in Table 4. All stability and transmission tests were conducted in an 8.0 m long, 2.3 m wide individual flume. Fig. 9 shows flume dimension, bottom slopes, wave gauge placement, and structure location for the hydraulic model tests. The covered beach sand compound slope was installed to represent local bathymetry seaward of the location of the geotextile tube structure. The cross section of the latter was placed on a horizontal section about 90 m (prototype distance) from the shoreline. Irregular waves were generated by a hydraulically actuated piston type wave machine. Wave heights were recorded by single wire capacitance type gauge. A total of twelve gauges were used during calibration of the facility, but only nine were used during the actual hydraulic model tests. Array No. 1 was positioned approximately 450 m prototype seaward of the geotextile tube toe location to obtain offshore wave heights. Array Nos. 2 and 3 provided the incident and transmitted wave data, respectively. The collected data were stored on a data acquisition system and analyzed using a computer analysis system. A schematic diagram of the laboratory equipment is shown in Fig. 10.

Photographs were taken prior to hydraulic model tests. The whole flume was flooded to the appropriate depth and the seabed was exposed to 1 cycle of low level waves. These initial waves were intended to allow settling and nesting of the newly constructed section as would occur under typical daily wave conditions prior to being exposed to relevant to our experimental

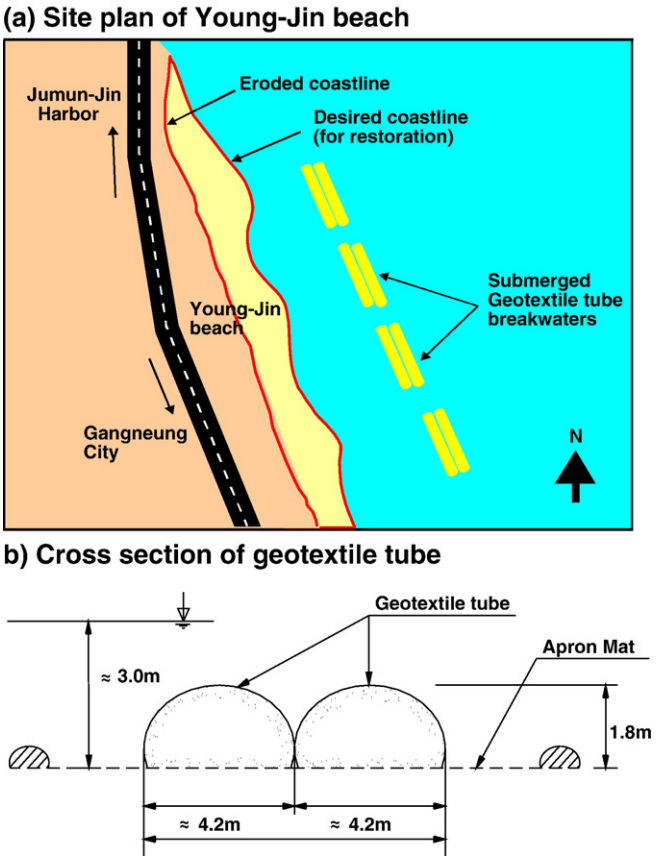


Fig. 16. Schematic diagram of geotextile tube at Young-Jin beach.

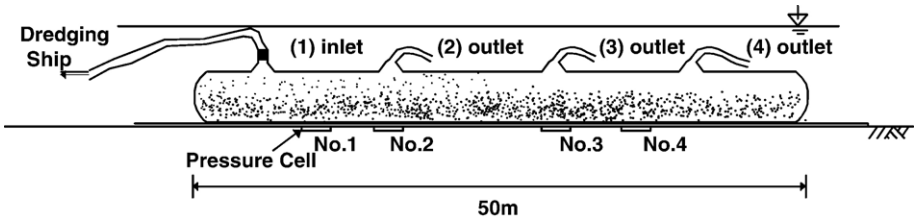


Fig. 17. Vertical stress measuring point.

design. The prototype duration for each wave height was 3 h. Test durations were completed with 1 cycle of random waves after 25 min. The response of the structure to each cycle of test waves was exhaustively recorded by video and photography.

4.4. Geotextile and fill material

The geotextiles most commonly used to construct geotextile tubes are either pure woven geotextile or are composites comprised usually of an external layer of woven geotextile and an internal layer of non-woven geotextile. The size and fabric specifications depend on the project requirements. Also, they should have a reasonable UV resistance. The interval of inlet is shorter for sandy soil while longer for the case of clayey soil (Leshchinsky, 1993). Physical properties of this geotextile are given in Table 5. The tensile strength test results of machine direction (MD) and cross machine direction (CD) are shown in Fig. 11.

The filling material of geotextile tube should be selected after consideration of the drainage capacity and filter properties, effective height after construction, and pumping equipments. One of the best filling materials is dredged material from near the construction site, which is effective for continuous filling process and more economical. In most cases, coarse grained material will dewater much faster than fine grained material such as silty clay. Hence sandy soil is effective for geotextile tube construction. In this study, the filling material was dredged sand obtained nearshore of Young-Jin beach. Filling and dredging processes were performed simultaneously. The physical properties of dredged filling materials are given in Table 6, and the grain-size distribution curve is shown in Fig. 12. The interface friction angle of dredged sand and geotextile was determined by a series of large-scale direct shear tests (ASTM D5321, 2002). The size of the geotextile used for the tests was 0.3 m × 0.3 m; the normal stress varied up to 700 kN/m^2. The interface friction angle is an important parameter in determining the stability of geotextile tubes when they are installed on sloping ground in, for example, shore protection projects such as the construction of breakwaters.

4.5. Results of hydraulic model tests

4.5.1. Stability

Stability analysis of coastal and hydraulic structure by hydraulic model testing was performed by measuring the displacement against wave attack. The significant wave height was increased in steps until the geotextile tube structure collapsed or until the highest obtainable significant wave height had been reached. The results of the hydraulic model test for stability are tabulated in Table 7. Based on these results, the two lines geotextile tube showed the lower displacement up to a significant wave height of 6.0 m. However, the single tube had already collapsed at 6.0 m significant wave height. Also, in the case of 0.5 H water depth above crest, damage by lower wave height occurred. From these results we conclude that the two lines geotextile tube and, for the case of zero water depth above crest is more stable than a single tube and 0.5 H water depth above crest against wave attack.

4.5.2. Wave transmittance properties

To evaluate the wave transmittance properties of a geotextile tube, the transmitted wave height was measured and transmitted wave spectrum analyzed. The wave transmission of the test sections could be calculated from the test results by Eq. (9).

$$R_{WTR} = \frac{H_T}{H_i} \tag{9}$$

in which R_{WTR} is the wave transmission coefficient, H_T is the transmitted wave height at the rear side of geotextile tube and H_i is the incident wave height (Shore Protection Manual, 1984). Full description of wave reduction by submerged structures can be found also in Pilarczyk (2003).

A total of 1024 transmitted wave height data were collected every 0.05 s. The incident and transmitted wave profiles are

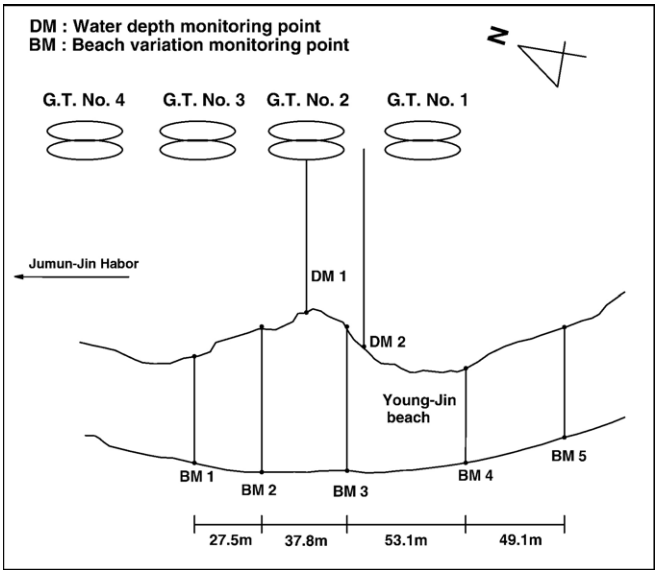


Fig. 18. Monitoring point layout.

(a) Barge and Crane



(b) Deploying the apron mat



(c) Placing the geotextile tube by crane



(d) Deploying the geotextile tube



(e) Connecting the injection nozzle and port



(f) Checking the vertical stress



(g) Checking the effective height



(h) Dredging equipment



Fig. 19. Steps in construction and placement of geotextile tube at Young-Jin beach.

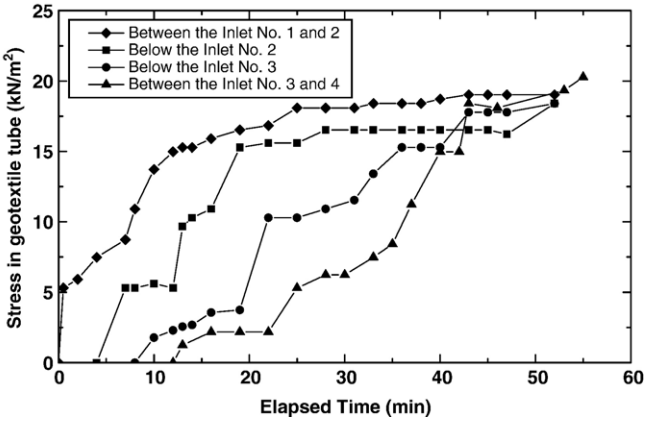


Fig. 20. Variation of vertical pressure with elapsed time.

shown in Figs. 13, 14 and 15) and the transmission coefficient is tabulated in Table 8. As the significant wave height increased, the transmitted wave height decreased due to interference by the geotextile tube structure. Also, two lines and zero water depth above crest case was more effective for wave adsorption. Based on the laboratory wave transmission test results, it should be noted that the relative width and crest height of the submerged structure are highly important factors influencing wave absorption properties.

5. The geotextile tube at Young-Jin beach

5.1. Design and construction

Based on the hydraulic model tests, we see that the two lines geotextile tube installed with zero water depth above the crest is more stable and more effective in wave absorption than other design plans. It was determined that two lines zero water depth above crest geotextile tube could be used at the Young-Jin beach. For shoreline protection the geotextile tube was installed 1.0 m below the water surface and as deep as possible, firstly to avoid spoiling the scenery and secondly, to reduce navigational risks. Fig. 16(a) shows a schematic diagram of the Young-Jin beach

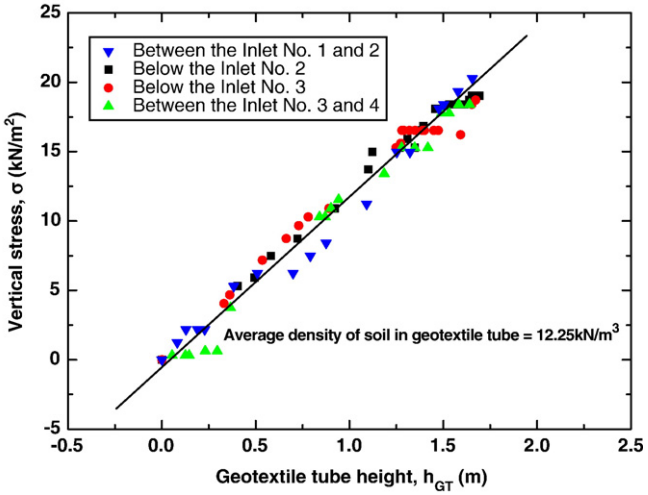


Fig. 21. Variation of vertical stress with effective height.



Fig. 22. Sea weed on the geotextile tube.

site. Geotextile tubes were designed two lines as detached breakwaters and had to be installed in about 3.0 m of water, 90–100 m distant from the shoreline. A single detached breakwater element implemented as a geotextile tube had a circumference 9.5 m (diameter 3.5 m), 50 m long and the effective height was 1.8 m (fill ratio 85%). It covered 240 m of near shore along the shoreline of Young-Jin beach. The cross section of the geotextile tube is shown in Fig. 16(b), the apron mat was installed as a fabric blanket to protect against scouring. The edge of the fabric was folded back 0.5 m and sewn forming a small tube that could be filled with sand. These small tubes help anchor the scour blanket. A cutter suction dredging ship was used to fill the geotextile tubes and the large barge ship (50 m long) was used in whole construction process. Also, special hardware (flanges, connectors, crane, etc) were required to connect the dredge discharge pipe to the tube.

5.2. Installation

A total of eight geotextile tubes were installed at the Young-Jin cove. The vertical stress at geotextile tube base and effective height were measured during and after construction. The shoreline variation and water depth of the offshore area were also monitored. Figs. 17 and 18 show the vertical stress measuring point and a plan view of the field monitoring scheme. Four of the two lines geotextile tubes were installed from south to north, spaced at 20.0 m intervals. The sequence of constructing a



Fig. 23. Sand accumulation by geotextile tube.

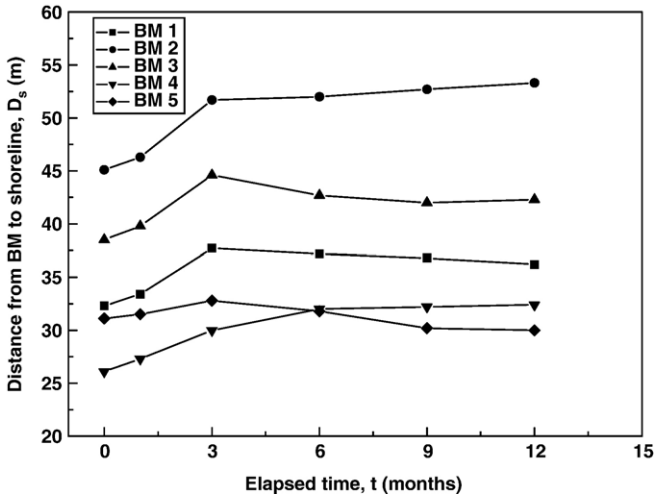


Fig. 24. Variation of shoreline with elapsed time.

geotextile tube can be summarized as follows: (1) survey the installation point, (2) deploy the apron mat, (3) place the geotextile tube, (4) connect the injection nozzle and port of the tube, (5) dredge and fill the tube with soil, (6) check the vertical stress and effective height, (7) complete the dredging and filling operation. This sequence is shown photographically in Fig. 19.

The soil used to fill the geotextile tube was hydraulically pumped by a cutter suction dredging ship. This dredge utilizes a 12.0 m digging cutter head, a 50 cm diameter discharge pipe, a 500 HP suction pump, and 20,000 lb torque horizontal auger to combine sediments and water into a slurry that is hydraulically transported to the tube via a 20 cm pipe. The maximum pump capacity of this dredging ship is about 500 m^3/h and the hydraulic pumping pressure was varied 14–22 kPa by pumping distance. Before and during the hydraulic filling, the pumping pressure, speed and mixing ratio of slurry were varied to determine the optimum value.

5.3. Observation of the filling process

The effective height and vertical pressure were monitored by the pressure cell at various time intervals during the filling process. The latter was conducted step by step by changing the inlet port. Fig. 20 shows the vertical pressure variation with elapsed time at the bottom of the geotextile tube. This figure indicates that the vertical pressure increased with filling time and that the settling and drainage occurred rapidly. As can be seen in Fig. 20, the construction time of one geotextile tube was less than 1 h and the desired final height was achieved after only four dredging and filling steps. Fig. 21 shows the relationship between vertical stress and effective height of the geotextile tube. The vertical stress increased on increasing the effective height, and the unit weight of dredged slurry estimated to be ca. 12.25 kN/m^3 from the relationship of vertical stress and effective height. As the dredging and filling process continued step by step, the effective height was increased and the filled sand inside the tube was dispersed from the inlet to the other outlet port.

5.4. Observation after construction

After the completion of geotextile tube construction, the four twin submerged detached geotextile tubes were successfully installed at different locations throughout the Young-Jin beach area. The size of a single geotextile tube is 50.0 m long and 1.8 m effective height. Figs. 22 and 23 show the submerged geotextile tube and the accumulation of sand during the 6 months following completion of construction. Sea weed has colonized the surface of the tube and sand has accumulated on its upper surface.

After the completion of the installation, the variations of shoreline and water depth near shore were monitored over a 12 month period. The variation of shoreline with the elapsed time is shown in Fig. 24. Based on the shoreline monitoring results, the shoreline at BM1 was extended by about 5.3 m during a 3 months period, because of short-term effectiveness of

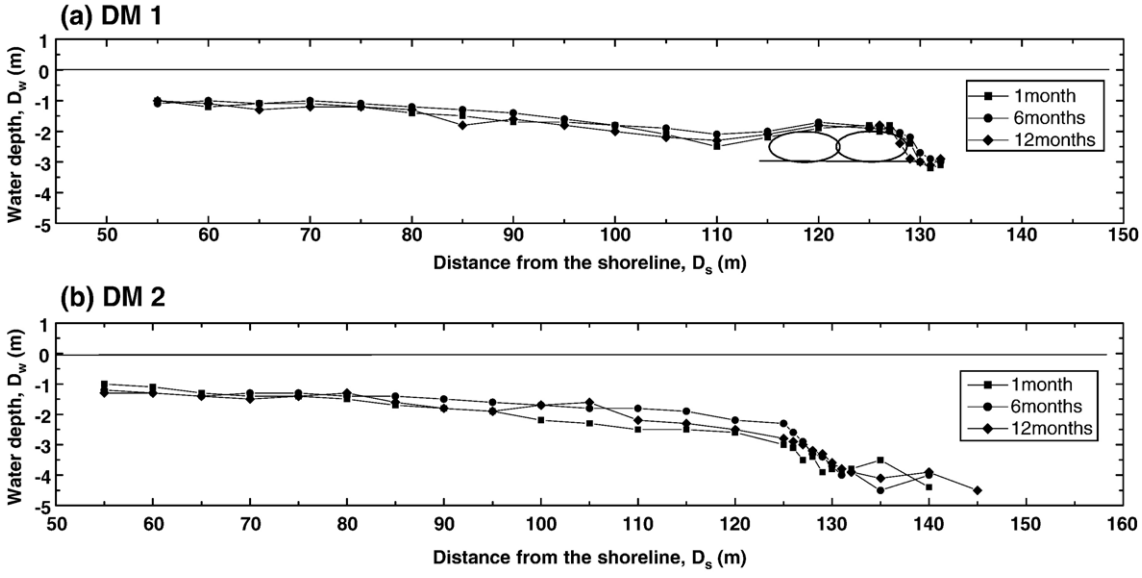


Fig. 25. Variation of water depth with elapsed time.

wave absorption and diminution of sea bed soil migration. However, after 3 months the shoreline of BM1 was re-eroded by tidal waves. We speculate that this arises from local effects (changed tidal flow direction and velocity) caused by the very geotextile installation itself. Also, the shorelines at BM3 and BM4 were monitored similarly to that at BM1, but the magnitude of re-erosion was relatively small. On the other hand, the shoreline of BM2 was continuously extended by the wave adsorption of tidal waves.

Fig. 25 shows the variation of water depth with elapsed time. The water depth was monitored at two different locations near shore (DM1, DM2), from the shoreline to the submerged geotextile tube. From Fig. 25, the water depth near shore increased with elapsed time, and the amount of sand accumulation around the geotextile tube covered area gradually increased up to 10.4 cm (Fig. 25(a)). On the other hand, at the geotextile tube situated on the sea side as well as between the tubes, a scouring effect caused by breaking waves and fast wave velocity, was developed (Fig. 25(b)).

6. Discussion and conclusions

This paper presents the case history of a geotextile tube construction on the east coast of Korea (Young-Jin beach). 2-D limit equilibrium analysis and hydraulic model tests of a geotextile tube at Young-Jin beach were conducted to select a suitable structural design. The construction procedures of geotextile tube and the results of in-situ measurement such as effective height, vertical stress of geotextile tube bottom, shoreline variation, and water depth in the near shore area were described. Based on the stability analysis, hydraulic model tests, and field measurements, the following discussion and conclusions can be derived:

To verify the feasibility of the 2-dimensional hydraulic stability analysis for submerged geotextile tube structures, this study compares theoretical calculations with hydraulic model test results. 2-D limit equilibrium stability analysis results indicate that the two lines tube is stable against overturning and bearing capacity failure, but the stability against sliding is high against significant wave heights of less than 3.0 m. Based on the hydraulic model tests, however, the two lines and zero water depth above crest case is stable against up to 5.0 m of significant wave height. As expected, the difference must be due to an inconsistency between the 2-dimensional limit equilibrium theory and the real behavior of the geotextile tube. In terms of real interactions between the geotextile tube and wave attack, there are two main inconsistencies. The first is the reduction of wave pressure caused by the elliptical and streamlined geotextile tube shape after filling. Secondly, the flexibility of a geotextile tube in the longitudinal direction. A 3-dimensional geotextile tube behaves, with regard to its flexibility, like a lumped mass of semi-infinite length. Thus, the comprehensive 2-dimensional hydraulic stability analysis deviates widely between theoretical safety factor and the behavior of real geotextile tube. However, in spite of inconsistencies in the 2-dimensional hydraulic stability analysis, it can still be used in stability analysis of geotextile tubes because it is very simple and easy to calculate.

Based on the laboratory wave transmission test results, the transmitted wave height decreased due to interference of geotextile tube structure. Further, the results show that the two lines, zero water depth above crest case is more effective for wave absorption. It should be noted that the relative width and crest height of the submerged structure has an important influence on wave adsorption capacity. From a comparison of field measurements and hydraulic model tests, the agreement of wave adsorption capacity between hydraulic model test data and the observed shoreline variation is quite good. Thus, from the results of hydraulic model tests, the shoreline variation after placement of a geotextile tube can be roughly predicted.

Observations of the filling process show that settling and drainage occurred very fast. The construction of one geotextile tube requires less than 1 h and the desired final height was achieved after only four dredging and filling steps. While the expeditious construction is most evident advantage of this innovative shore protection technology, the favorable economic aspect should not be overlooked.

After 1 year in use, seaweed had inhabited the surface of the submerged tube. Hence, we conclude that the polymer material used in their manufacture is unlikely to have an adverse effect on marine life: Moreover, it can be environmentally sustainable to the adjacent ecology. The long-term effectiveness of the installations however, cannot be judged until more time has passed. The test sites have been exposed to a number of severe storms common to the east coast of Korea. This makes it difficult to evaluate the structure's performance as a form of innovative shore protection. Over the short period of our observations, the variation of shoreline with the elapsed time was extended into the sea during 3 months, because of short-term effectiveness of wave adsorption and decrement of sea bed soil migration. After 3 months, however, the shoreline was re-eroded by tidal waves and geometrical reason. However, the magnitude of re-erosion was relatively small compared to the extension of the shoreline. The water depth in the near shore area decreased with elapsed time, and the sand gradually accumulated around areas covered by the geotextile tube.

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